

Molecular pathways to neurodegeneration

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The molecular bases underlying the pathogenesis of neurodegenerative diseases are gradually being disclosed. One problem that investigators face is distinguishing primary from secondary events. Rare, inherited mutations causing familial forms of these disorders have provided important insights into the molecular networks implicated in disease pathogenesis. Increasing evidence indicates that accumulation of aberrant or misfolded proteins, protofibril formation, ubiquitin-proteasome system dysfunction, excitotoxic insult, oxidative and nitrosative stress, mitochondrial injury, synaptic failure, altered metal homeostasis and failure of axonal and dendritic transport represent unifying events in many slowly progressive neurodegenerative disorders.

Evidence is accumulating to suggest that such chronic neurodegenerative disorders as Alzheimer's disease (AD), Parkinson's disease (PD), Huntington's disease (HD) and amyotrophic lateral sclerosis (ALS) are caused by a combination of events that impair normal neuronal function. Clinical signs are evident before frank neuronal loss. Therefore, new efforts have focused on identifying crucial changes, of genetic, epigenetic, or environmental origin, that hamper normal neuronal function.

In this review, we summarize the current understanding of molecular pathways implicated in common neurodegenerative disorders, trying to clarify similarities that might provide a foundation for their treatment.

Molecular pathways to AD

AD is the most common neurodegenerative disorder worldwide. Approximately 4.5 million people suffer from this devastating condition in the United States alone. In AD, neurons of the hippocampus and cerebral cortex are selectively lost. Brains of individuals with AD manifest two characteristic lesions: extracellular amyloid (or senile) plaques and intracellular neurofibrillary tangles of hyperphosphorylated tau protein (Fig. 1a)¹. Amyloid plaques contain small, toxic cleavage products (denoted as A β 40 and A β 42) of the amyloid precursor protein (APP). The apoE4 (apolipoprotein E4) genotype is a powerful risk factor for developing AD, and it may possibly affect A β deposition and neurofibrillary tangle formation².

Mutations in three genes that are inherited in an autosomal dominant fashion have been linked to rare familial, early-onset forms of AD. These genes include those encoding APP, presenilin 1 (PS1) and presenilin 2 (PS2). Although these familial forms account for only a few cases of AD, one common event in both familial and sporadic types of AD is the increased production and accumulation of the toxic A β . This observation led to the 'amyloid cascade hypothesis' that excessive A β production is the primary cause of the disease^{3,4}.

Do plaques and tangles cause AD, or are they simply the telltale remains of earlier, pivotal events that led to the disease? One problem has been that in people with AD the density of amyloid plaques correlates poorly with the severity of dementia³. Furthermore, although neurofibrillary tangles correlate well with decline in cognitive skills, they seem to be a late event and in some cases possibly downstream of A β accumulation⁵. Recent findings, however, indicate that dendritic and synaptic injury occur early and that protofibrils and oligomers of A β 40 and A β 42, rather than large plaques, cause neuronal dysfunction.

APP is a type I membrane protein and contains a large extracellular region, a transmembrane helix and a short cytoplasmic tail (Fig. 1b). The N-terminal half of APP contains a heparin-binding domain (N-APP), a copper-binding domain (CuBD) and an APP protease inhibitor domain (APPI) (Fig. 1b). The N-APP structure reveals that APP may belong to a superfamily of cysteine-rich growth factors with a putative heparin-binding site⁶. The CuBD structure is homologous to copper chaperones. This domain may regulate dimerization or proteolytic processing, or it may act as a metal transporter⁷. There is also structural information on the A β peptides that originate by proteolytic processing of APP^{8,9}. Both A β 40 and A β 42 adopt variable and partly helical structures dependent on membrane binding, metal chelation and interaction with other peptides. A β binds Cu²⁺, Fe²⁺ and Zn²⁺ coordinated by three histidines (His677, His684, His685) and a tyrosine (Tyr681). Metal binding induces a β -sheet-like conformational change in A β , resulting in enhanced aggregation.

Toxic A β originates from regulated intramembrane proteolysis of APP by a complex of secretases (Fig. 1c). The first cleavage of APP is mediated by β - or α -secretase, releasing most of the extracellular portion of APP as two fragments, APPs- α and APPs- β , leaving behind the C-terminal membrane bound fragment (Fig. 1c). This portion of APP is then cleaved by a large protein complex, γ -secretase, at several sites including amino acid (aa) 711 (A β 40) and at least three additional subsites at aa713 (A β 42), aa714 (A β 43) and aa720 (A β 49) (Fig. 1b). Several mutations in APP, such as the Swedish mutation, cluster at the β -secretase cleavage sites; these mutations result in increased amounts of A β peptide and protofibril formation¹⁰.

The precise composition of the γ -secretase complex is still under debate, but PS1, nicastrin, Aph-1 and Pen-2 seem to be required^{11–13}.

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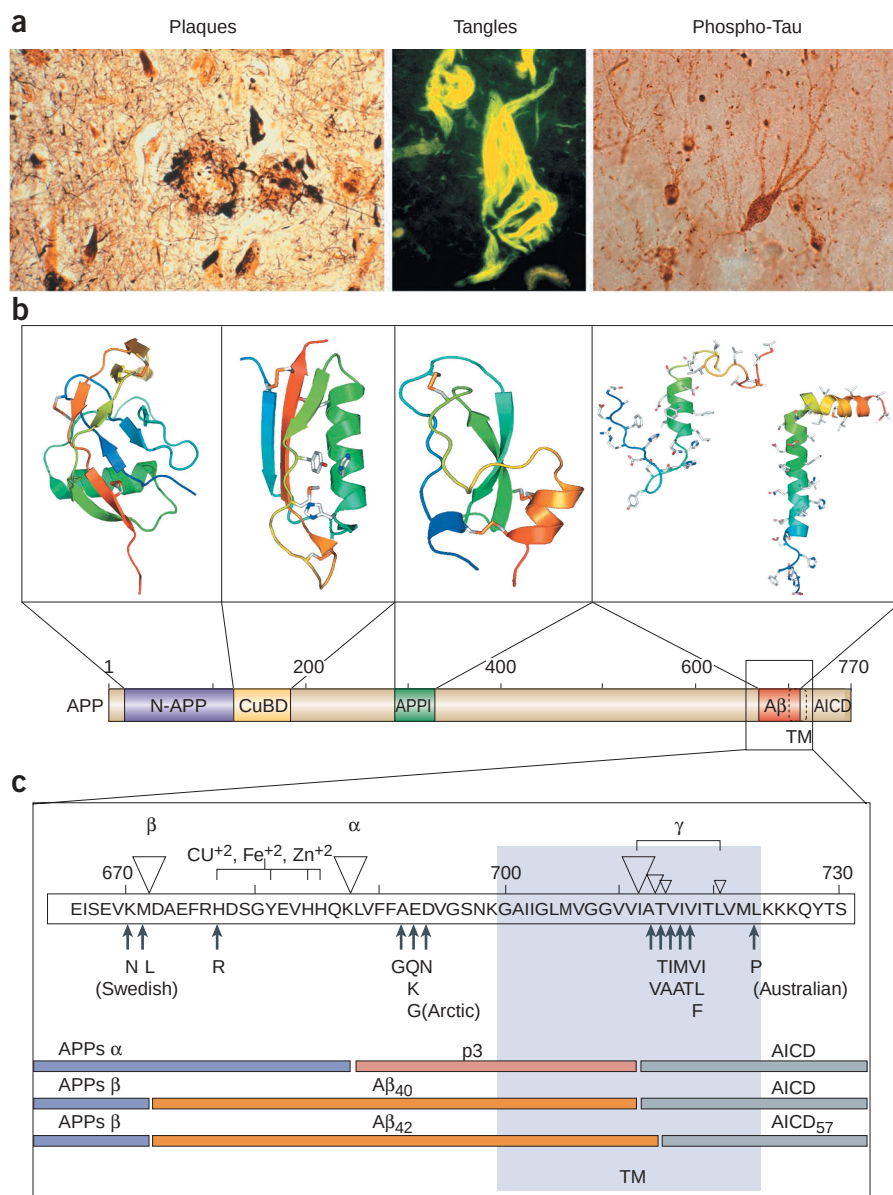


Figure 1 Histology and structures related to AD. **(a)** Silver-stained amyloid plaques, neurofibrillary tangles and immunocytochemistry of phosphorylated tau in brain sections from AD patients. **(b)** Structures and domain model of amyloid β 4 precursor protein (APP, PDB entry P05067; N-terminal growth factor domain (N-APP), aa 28–123, PDB entry 1MWV; APP copper-binding domain (CuBD), aa 124–188, PDB entry 1OWT; APP protease inhibitor domain (APPI), aa 287–344, PDB entry 1AAP; amyloid β peptide (A β 42), PDB entry 1IYT; A β 40), PDB entry 1BA6; transmembrane domain (TM). Structural models are shown in ribbon representation, color-coded from N terminus (blue) to C terminus (red). Side chains of relevant residues are shown as ball-and-stick figures. **(c)** APP residues 665–730; the TM domain (residues 700–723) is highlighted in light blue; cleavage sites of secretases α , β and γ are indicated, along with metal-coordinating residues (His677, Tyr681, His684, His685) and mutations associated with AD (<http://molgen-www.uia.ac.be/ADMutations/>). Proteolytic processing and resulting fragments of APP: APPs α aa 1–686; APPs β , aa 1–670; p3 peptides, aa 687–711 or 713; A β 40, aa 671–711; A β 42, aa 671–713. APP intracellular domain (AICD), aa 711 or 713–770.

(NMDA) subtype results in increased intracellular Ca^{2+} , which activates neuronal nitric oxide synthase and consequently generates nitric oxide (NO). When generated in excess, NO combines with superoxide anion (O_2^-), forming the highly reactive and neurotoxic product peroxynitrite (ONOO^-), which leads to further oxidative and nitrosative stress in part via mitochondrial injury. In fact, positive phase III human trials of the uncompetitive NMDA receptor channel blocker, memantine, led to its recent approval for the treatment of AD²². A third view suggests that protofibrils and aggregates convey harmful effects to neurons by paralyzing axonal and dendritic

PS1 is a transmembrane domain aspartyl protease that cleaves its substrates in the membrane-spanning region. PS1 is probably responsible for the generation of A β fragments.

More than 100 missense mutations in PS1 and PS2 have been identified in rare familial, early-onset AD¹⁴. Experiments in culture and transgenic mice reveal that these mutations result in increased A β production^{15,16}. Conversely, mice lacking PS1 have decreased A β 40 and A β 42 production^{17,18}, suggesting that PS1 has a pivotal role in γ -secretase activity. C-terminal cleavage of APP by caspase enzymes may also be required for toxicity¹⁹.

It is still unclear how A β does its damage, but several mechanisms have been proposed. One view suggests that A β protofibrils activate microglia, inciting an inflammatory response and release of neurotoxic cytokines. Nonsteroidal anti-inflammatory drugs (NSAIDs) including ibuprofen seem to delay the onset of AD²⁰. Additionally, NSAIDs reduce the production of A β 42 (ref. 21). In a second view, A β protofibrils trigger excessive release of excitatory amino acids like glutamate from glial cells that may injure nearby neurons by excitotoxicity. Overactivation of glutamate receptors of the *N*-methyl-D-aspartate

transport. Both APP and PS may bind kinesin I and regulate vesicular traffic^{23–25}. PS mutants increase glycogen synthase kinase-3 β activity, which hampers kinesin-mediated, anterograde axonal transport²⁵. Also, A β deposits may act as nonspecific ‘roadblocks’, representing a physical transport barrier.

An additional mechanism of A β injury is synaptic dysfunction and loss, which are early events in AD and occur before amyloid plaque formation²⁶. Cholinergic transmission and synaptic density are considerably decreased in AD patients. The mechanism for synaptic damage is unknown, but diffusible oligomeric forms of A β may be important. Synaptic dysfunction probably contributes to memory loss and cognitive deficits in AD. In fact, APP transgenic mice manifest cellular, biochemical and electrophysiological evidence of synaptic deficits before A β deposition, including reduced excitatory postsynaptic potentials and LONG-TERM POTENTIATION (LTP), regarded as a correlate of learning and memory²⁷. Inhibition of γ -secretase decreases oligomeric A β and LTP deficits²⁸. Microinjection of A β 43 and A β 40 peptides into the rat hippocampus disrupts synaptic transmission and short-term memory²⁹.

A β may also mediate harmful effects by binding redox-reactive metals, which in turn release free radicals^{30–36}. We found that nitrosative stress mobilizes zinc from intracellular stores, resulting in mitochondrial dysfunction³⁷. Chelation of zinc and copper provides neuroprotective effects³⁸. For example, clioquinol (CQ), an antibiotic that also chelates zinc and copper and crosses the blood-brain barrier, decreases brain A β deposition and improves learning in mutant APP transgenic mice³⁹. A recent human phase II clinical trial of CQ for AD seems promising⁴⁰.

Oxidative stress from mitochondrial dysfunction occurs early in AD, and A β may directly or indirectly injure mitochondria^{41,42}. A β blocks respiratory complex I, thus producing a decline in ATP⁴³. In isolated mitochondria, A β inhibits respiration and enzyme activity (α -ketoglutarate dehydrogenase and pyruvate dehydrogenase)⁴⁴. The mechanism of A β translocation to mitochondria remains unknown. New data describe an interaction partner of A β in mitochondria, termed A β -binding alcohol dehydrogenase (ABAD)⁴⁵. ABAD is upreg-

ulated in neurons of AD patients. Expression of ABAD in concert with mutant APP enhances free-radical production and toxicity. Conversely, a peptide that blocks A β -ABAD interaction prevents free-radical formation and cell death. A β interacts with the LD loop of ABAD, close to its nicotinamide adenine dinucleotide (NAD) binding site, and may therefore induce a conformational change that prevents NAD binding. To date, however, details of the interaction are not apparent in the crystal structure⁴⁵.

In summary, mitochondrial dysfunction and resulting energy deficits may contribute to impaired clearance of protein aggregates and neuronal dysfunction, affecting ion channel and pump activity, neurotransmission, and axonal and dendritic transport.

Molecular pathways to PD

PD is the most common neurodegenerative movement disorder. Approximately 1% of the population older than 65 years suffers from this slowly progressive neurodegenerative disease; 95% of PD cases are

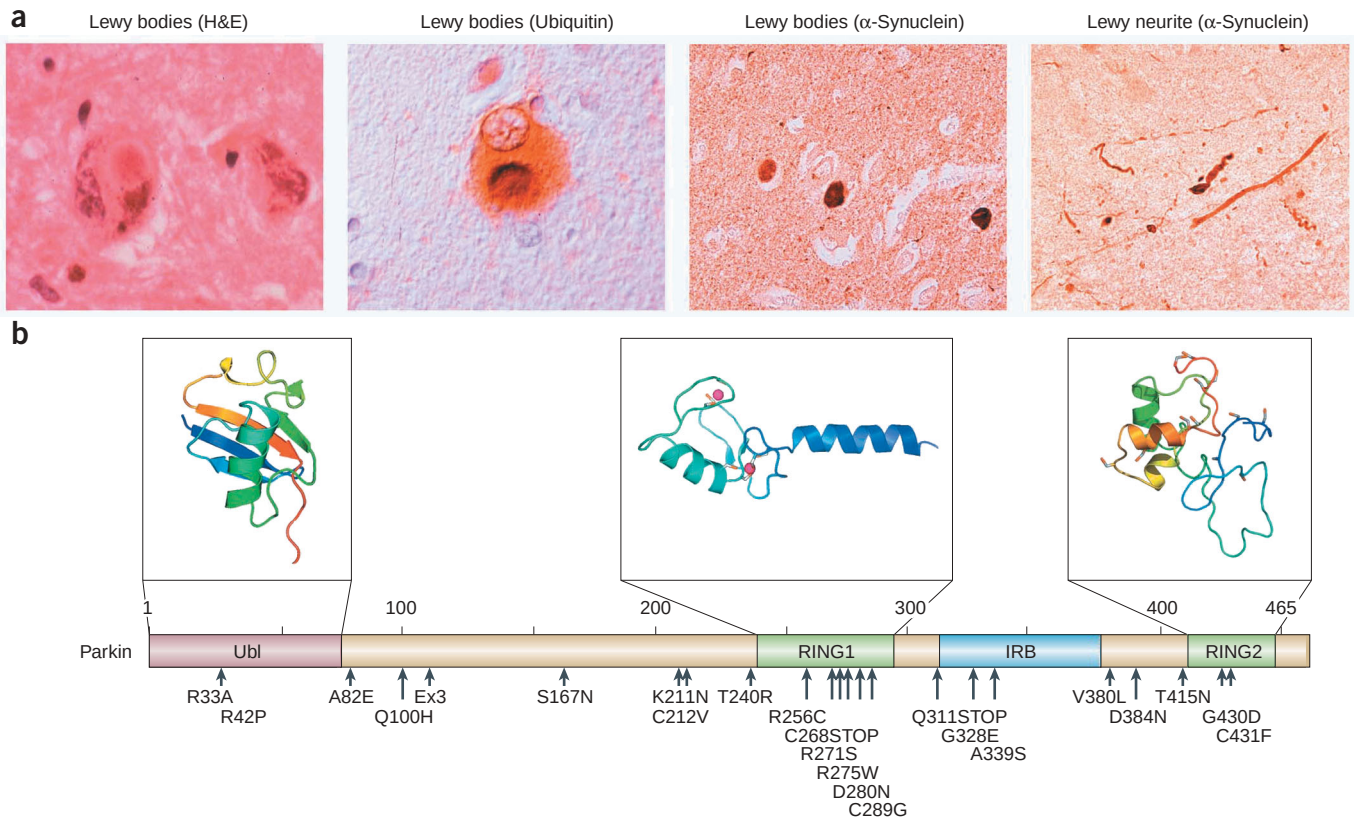


Figure 2 Histology and structures related to PD. **(a)** Lewy bodies and neurites visualized by H&E staining or immunocytochemistry using antibodies specific for ubiquitin and α -synuclein in brain sections of sporadic cases of PD. **(b)** Structures and domain model of human parkin (NCBI AB009973); N-terminal ubiquitin-like domain (Ubl), aa 1–79, PDB entry 1IYF; homology model of RING1 domain, aa 238–293 based on PDB entry 1JM7; IRB domain, aa 314–377; homology model of RING2 domain, aa 418–449, based on PDB entry 1FO4. Structural models are depicted in ribbon representation, color-coded from N terminus (blue) to C terminus (red). Cysteine side chains located in the RING zinc finger domains are depicted in ball and stick with Zn²⁺ as a pink sphere. Arrows indicate reported familial-associated Parkinson's disease-causing mutations. **(c)** Schematic representation of parkin interactions. The Ubl domain binds to the proteasome. Substrate binding may be mediated by the RING1 domain and E2 enzyme binding by the IRB and RING2 domains.

GLOSSARY

Long-term potentiation (LTP) An enduring increase in the amplitude of excitatory postsynaptic potentials as a result of high-frequency (tetanic) stimulation of afferent pathways. It is measured both as the amplitude of excitatory postsynaptic potentials and as the magnitude of the postsynaptic-cell population spike. LTP is most often studied in the hippocampus and is often considered to be the cellular basis of learning and memory in vertebrates.

Haploinsufficiency Loss of one copy (one allele) of a gene is sufficient to give rise to disease. Haploinsufficiency implies that no dominant-negative effect of the mutated gene product has to be invoked.

Dominant negative A mutant molecule that in some cases can form a heteromeric complex with the normal molecule, knocking out the activity of the entire complex.

Small interfering RNA RNA that can be used to interfere with normal RNA processing, causing rapid degradation of the endogenous RNA and thereby precluding translation.

Missense mutation A mutation that results in the substitution of an amino acid in a protein, which therefore often has abnormal function.

Knock-in transgenesis The insertion of a mutant gene at the exact site of the genome where the corresponding wild-type gene is located. This approach is used to ensure that the effect of the mutant gene is not affected by the activity of the endogenous allele.

sporadic. The symptoms of PD are caused by selective and progressive degeneration of pigmented dopaminergic (DA) neurons in the substantia nigra pars compacta. Current treatments, such as administration of L-DOPA to produce dopamine, are only symptomatic and do not stop or delay the progressive loss of neurons. In fact, some studies have suggested that oxidative injury via dopamine may lead to further neuronal damage⁴⁶.

One important feature of PD is the presence of eosinophilic, cytoplasmic inclusions of fibrillar, misfolded proteins, termed Lewy bodies, in affected brain areas (Fig. 2a). The exact composition of Lewy bodies is unknown, but they contain ubiquitinated α -synuclein, parkin, synphilin, neurofilaments and synaptic vesicle proteins (Fig. 2a).

The etiology of PD remains unclear⁴⁷. Hints that sporadic PD can be initiated by environmental toxins are provided by the accidental discovery that 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP), an herbicide and contaminant of illicit street drugs, causes parkinsonian-like symptoms in human substance abusers and in animal models^{48–52}. Astrocytes metabolize MPTP to 1-methyl-4-phenyl pyridium (MPP⁺) via monoamine oxidase B. MPP⁺ is selectively imported into DA neurons via the dopamine transporter, where it targets mitochondria, inhibits respiratory complex I and promotes reactive oxygen species production. As with AD, altered metal homeostasis may have a role in the etiology of PD. Metal chelation by CQ rescues DA neurons and improves balance and posture in MPTP-treated animals⁵³.

Additional support for mitochondrial dysfunction in PD pathogenesis comes from evidence that the insecticide rotenone induces a parkinsonian-like syndrome in animal models and probably humans, with protein deposits that resemble Lewy bodies⁵⁴. Rotenone also inhibits mitochondrial complex I, and DA neurons seem to be most severely affected.

Rare hereditary forms of PD have provided insight into the molecular pathways of this disorder⁴⁷. Mutations in at least four genes have been linked to PD, including α -synuclein (*PARK1*), parkin (*PARK2*), DJ-1 (*PARK7*), and PTEN (phosphatase and tensin homolog deleted on chromosome 10)-induced kinase 1 (*PINK1*, also known as *PARK6*)^{55–58}.

α -Synuclein mutations are autosomal dominant and represent toxic gain-of-function mutations, resulting in abnormal protein accumulations^{59,60}. The normal function of α -synuclein is unknown.

The protein is normally unfolded, revealing little three-dimensional structure. α -synuclein can polymerize into filaments and is found in Lewy bodies and neurites (Fig. 2a). The molecular pathways of α -synuclein-mediated toxicity are unknown; however, increased oxidative stress, mitochondrial injury and altered cellular transport have been proposed.

Mutations in parkin are found in juvenile PD and are inherited as autosomal recessive, causing a loss in parkin function. New findings indicate that parkin mutations may be linked not only to juvenile PD, but also to late-onset cases of PD. Heterozygous mutations in parkin may also cause PD, indicating a dominant form of inheritance in some cases. Perhaps HAPLOINSUFFICIENCY causes disease when coupled with nitrosative and oxidative stress (see later). Alternatively, some mutations may be DOMINANT NEGATIVE in nature.

Parkin is an E3 ligase, catalyzing the addition of ubiquitin to specific substrates that targets them for degradation by the ubiquitin-proteasome system (UPS). The parkin protein has several domains (Fig. 2b). Structural information for the N-terminal ubiquitin-like (Ubl) domain (residues 1–75) reveals that it binds to the Rpn10 subunit of the 26S proteasome. The Arg42Pro mutation, found in juvenile PD, is close to the Rpn10 subunit binding site, possibly impairing its proteasomal interaction⁶¹. The C-terminal half of parkin contains two RING domains, common to several E3 ligases, connected by an in-between RING domain (IBR). The RING domains are cysteine-rich zinc fingers implicated in substrate recognition and binding to E2 enzymes that transfer ubiquitin⁶². The RING1 domain constitutes a hotspot for missense mutations. Two mutations located in RING1, Arg256Cys and Arg275Trp, cause altered protein localization and increased aggregation⁶³. Most mutations clustered in the conserved domains of parkin diminish enzyme activity, suggesting that parkin loss of function causes the disorder (Fig. 2b).

Parkin substrates include glycosylated α -synuclein, synphilin, CDCrel-1 and the Pael-R receptor. These substrates may accumulate when parkin E3 ligase function is lost by mutation or other mechanism, contributing to neurotoxicity. Overexpression of wild-type parkin improves cell survival after various insults in a number of systems^{64–67}. Parkin blocks α -synuclein- and Pael-R-mediated toxicity, probably by promoting their degradation^{64,65,68}. Parkin also mediates the clearance of some polyglutamine repeat proteins⁶⁹. In *Drosophila*,

parkin protects cells from mitochondrial death pathways⁷⁰. Hence, enhancing parkin activity may be therapeutic.

New findings, however, suggest that parkin may have other functions besides targeting proteins for degradation via the UPS. For instance, parkin knockout mice do not manifest accumulation of parkin substrates, and humans with parkin mutations lack Lewy bodies. These data imply that parkin loss of function may cause degeneration by a mechanism other than aberrant protein accumulation. Parkin knockout mice manifest decreased oxidative phosphorylation and increased oxidative stress⁷¹. Parkin may be localized to mitochondria, and its loss induces abnormal mitochondrial morphology, muscle degeneration and impaired spermatogenesis in *Drosophila*⁷⁰.

Additionally, parkin is a target of oxidative and nitrosative stress in sporadic PD. Cysteine residues in the RING domains are sensitive to nitrosative and oxidative modifications, which alter protein function. New findings indicate that parkin's E3 ligase activity is modified by NO, thus linking environmental stress to a molecular abnormality and a clinical phenotype similar to that seen in hereditary forms of PD^{72,73}.

A third gene linked to recessively inherited albeit rare PD is *DJ-1* (*PARK7*)⁵⁷. In a Dutch kindred, a large homozygous deletion of *DJ-1*

leads to protein loss, whereas in an Italian family, a Lys166Pro point mutation occurs that results in destabilization of the protein. DJ-1 is implicated in multiple processes, including oncogenic transformation, gene expression, mRNA sorting and chaperone activity^{74,75}. DJ-1 mediates an oxidative stress response. In cells treated with the mitochondrial poison paraquat or MPP⁺, DJ-1 becomes more acidic. DJ-1 can scavenge H₂O₂, whereas the Lys166Pro mutant is inactive. DJ-1 knockdown by SMALL INTERFERING RNA sensitizes cells to oxidative stress. The crystal structure of DJ-1 (refs. 76,77) reveals a flavodoxin-like fold similar to PH1704 protease and heat shock protein Hsp31 (Fig. 3). DJ-1 forms a dimer, which is destabilized by the Lys166Pro mutation at the dimer interface. DJ-1 does not contain the typical active-site triad expected for a cysteine protease. Cys106, located in the putative active site, and Cys56, which is located on the surface and forms a disulfide bond with its dimer counterpart, are potential sites for oxidative and nitrosative stress, which could modify DJ-1 function in sporadic PD.

New data reveal that mutations in the gene encoding PTEN-induced kinase 1 (PINK1) are linked to early-onset PD⁵⁸. Normal and mutant PINK1 localize to mitochondria. PINK1 contains a highly conserved kinase domain similar to serine/threonine kinases in the Ca²⁺-calmodulin family. Two mutations associated with PD reside in this domain, and homology modeling reveals that the MISSENSE MUTATION Gly309Asp is located in its ADP binding site, probably interfering with ADP binding and kinase activity (Fig. 3b). The Trp437STOP mutation causes truncation of PINK1 in the helical kinase domain, probably destroying the enzyme (Fig. 3b). Expression of normal PINK1 protects cell lines from loss of mitochondrial membrane potential and apoptosis induced by proteasome inhibitors, an effect lost with the PINK1 mutant Glu309Asp (ref. 58). These findings suggest that PINK1 may phosphorylate a mitochondrial target(s) to protect against cellular stress.

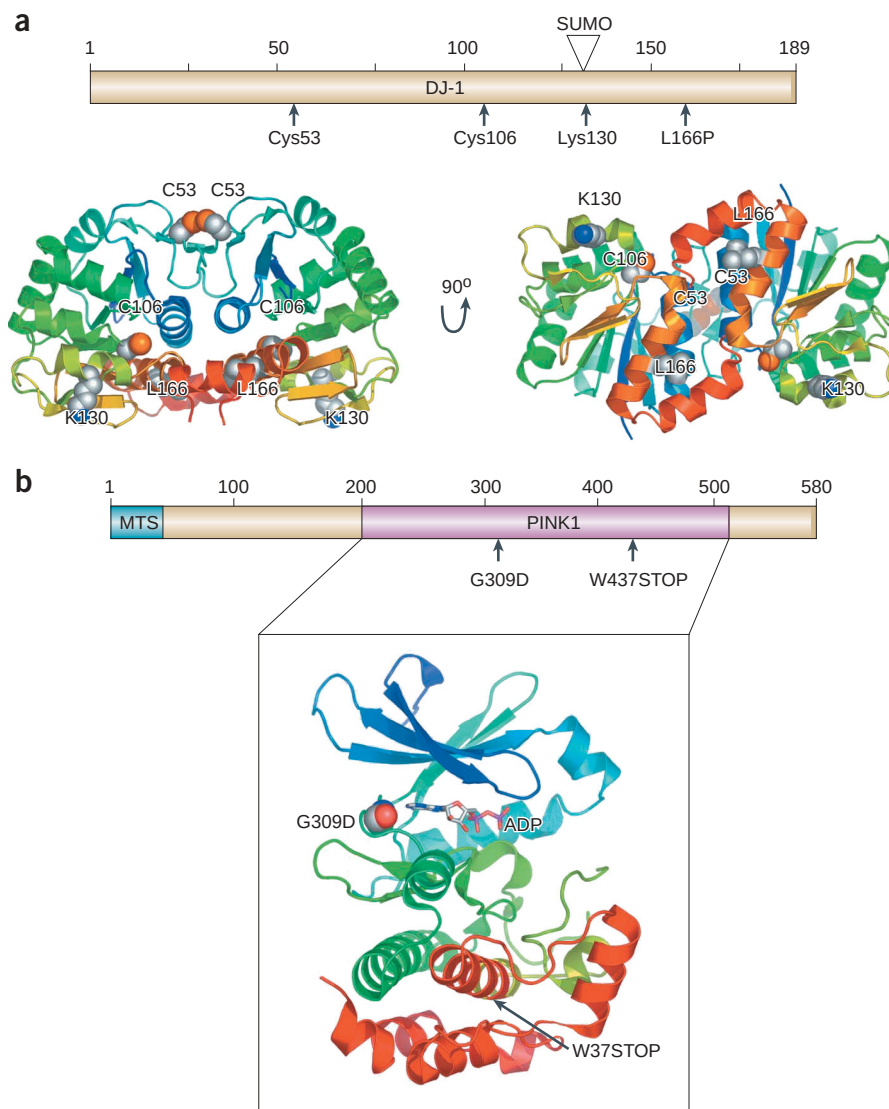


Figure 3 DJ-1 and PINK1 proteins in PD. (a) Domain scheme and crystal structure of human DJ-1. Top: Domain structure of DJ-1 with relevant residues indicated. Bottom: Ribbon representation of human DJ-1 structure (PDB entry 1PEO), color-coded from N terminus (blue) to C terminus (red). Side chains for two cysteine residues, Cys53 and Cys106, implicated in the oxidative stress response, for Lys130, which serves as a sumoylation site, and for Leu166, which is mutated to proline, are depicted in cpk mode. (b) Top: Domain scheme of human PINK1 with the mitochondrial targeting sequence (MTS, aa 1–34) shown in blue and the conserved kinase domain (aa 200–520) shown in pink. Location of the mutations G309D and W437STOP, linked to hereditary early-onset PD, are indicated (arrows). Bottom: Homology model of the PINK1-kinase domain (aa 205–510) based on PDB entry 1MQ4 shown in ribbon representation, color-coded from N terminus (blue) to C terminus (red). An ADP molecule is depicted in ball-and-stick mode. Locations are shown for mutations G309D (depicted in cpk mode) and W437STOP (arrow).

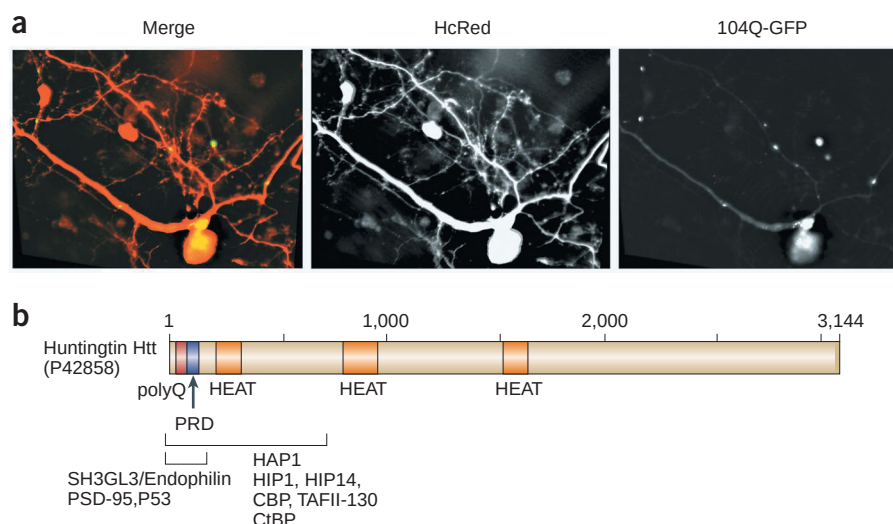


Figure 4 Aggregation and domain structure of Htt in HD. (a) Aggregates of polyQ repeats in dendrites and nuclei in a primary neuron co-transfected with vectors encoding 104Q-GFP and HcRed (Clontech). (b) Huntingtin (Htt) domain model indicating the location of the polyQ repeat, proline-rich domain (PRD) and HEAT repeats. Selected interaction partners are listed for the polyQ and PRD region as well as the N-terminal region, including the first HEAT repeat section. See text and references for abbreviations.

Pathways to HD

HD is an autosomal dominant inherited neurodegenerative disorder affecting 1 in 10,000 individuals. It is caused by an insertion of multiple CAG repeats in the huntingtin gene. This results in an N-terminal polyglutamine (polyQ) expansion of the large protein huntingtin (Htt), similar to other polyQ-related neurodegenerative disorders. Disease severity depends on the length of the polyQ stretch, with repeats ≥ 40 clearly linked to HD. The polyQ expansion is thought to confer a toxic gain of function with selective loss of neurons in the striatum and cerebral cortex.

Htt expression is ubiquitous but is greatest in neurons. Htt is reported in membranes, cytosol, nuclei, and mitochondria, and in association with microtubules^{78–80}. The expanded polyQ stretch in mutant Htt induces a conformational change resulting in aggregates in dendrites and nuclei (Fig. 4a). It is not clear whether aggregates, smaller accumulations or soluble aberrant protein are important for HD pathogenesis.

The function of normal Htt and the mechanism whereby mutant Htt mediates harmful effects remain unclear⁸¹. Htt may act as a molecular scaffold, regulating several cellular processes including endocytosis, vesicle transport, excitatory synapses, transcriptional events and mitochondrial function^{82,83}.

The Htt domain model reveals a polyQ and proline-rich region at the N terminus followed by three HEAT domains, named for the four proteins in which they were first detected: Htt, elongation factor 3, the regulatory A subunit of protein phosphatase 2A, and TOR1 (ref. 84) (Fig. 4b).

Several proteins interact with the Htt proline-rich domain. An example is endophilin 3, which regulates dynamin and synaptotagmin function⁸⁵. Htt also binds to postsynaptic density protein-95 (PSD-95), a molecule that interacts with excitatory aa receptors, including NMDA receptors, and synaptic GTPase-activating protein (SynGAP)⁸⁶. SynGAP in turn inhibits Ras and thereby regulates excitatory synapses. Thus, Htt may modulate synaptic transmission, learning and memory via its interaction with PSD-95.

HEAT domains mediate protein-protein interactions. Several proteins have been isolated that bind to the HEAT repeat. Among them are Htt-interacting protein (HIP)-1, HIP14 and Htt-associated protein (HAP1), which have roles in endocytosis and membrane trafficking^{87–90}. Impaired axonal and dendritic trafficking are also implicated in HD⁹¹, and transgenic KNOCK-IN mice for mutant Htt 150Q manifest

axonal pathology suggestive of transport defects⁹². Similarly, expression of mutant Htt in *Drosophila* produces an axonal transport phenotype⁹³.

Mutant Htt may be neurotoxic via transcriptional dysregulation, binding and sequestering selective transcription factors and coactivators. These include cAMP response element binding protein (CREB) binding protein (CBP), p53, co-repressor C-terminal binding protein (CtBP), Sp1 and TAFII-130 (refs. 94–97). Histone deacetylase inhibitors, which increase transcription, ameliorate polyQ-mediated neurodegeneration in *Drosophila*⁹⁴.

Mitochondrial dysfunction is also implicated in HD pathogenesis. The mitochondrial toxin 3-nitropropionate produces neuropathology in animals similar to that observed in human HD^{98,99}. Mitochondrial membrane potential in lymphocytes of HD patients is depolarized at lower calcium concentrations than normal⁸⁰, and brain mitochondria isolated from transgenic mice expressing full-length mutant Htt show similar deficiencies in calcium handling. These findings suggest that mutant Htt may make neurons susceptible to excitotoxicity, associated with increases in cytosolic calcium.

Molecular pathways to ALS

ALS (or Lou Gehrig's disease) involves degeneration of motor neurons, resulting in progressive muscle wasting and weakness, culminating in paralysis, respiratory failure and death. Perhaps 10% of cases are familial, and of those, ~2–3% are caused by mutations in the gene encoding Cu/Zn superoxide dismutase-1 (SOD1), producing a toxic gain of function rather than loss of (catalytic) function¹⁰⁰. The precise pathogenic mechanism is not clear, but implicated in motor neuron dysfunction and death are protein misfolding and aggregation, defective axonal transport, mitochondrial dysfunction and excitotoxicity via faulty glutamate reuptake into glial cells. Recent structural evidence suggests that some Cu/Zn SOD1 mutations result in destabilization of normal dimers of the enzyme and foster aggregation, forming amyloid or pores depending on the conditions, not unlike familial amyloid polyneuropathy^{101,102}. Stabilization of dimers has therefore been proposed as a therapeutic intervention¹⁰³.

Recent findings in SOD1 mutant mice indicate that motor neuron death is not cell autonomous and depends on surrounding cells. Neurons that express mutant SOD1 protein can be rescued by nearby non-neuronal wild-type cells¹⁰⁴.

A recent report on sporadic ALS (representing the vast majority of cases) revealed abnormal RNA editing in GluR2 subunits of glutamate receptors, producing increased Ca^{2+} entry into neurons^{105,106}. This mechanism may contribute to neuronal demise, suggesting possible therapeutic targets, such as counteracting overly active calcium-permeable glutamate receptors or compensating for potentially dysfunctional RNA-editing enzymes.

Conclusions

Here we review common themes occurring in several neurodegenerative disorders. Slowly progressive neurodegenerative diseases are probably not the result of a single hit-and-run event, but rather a several-step process involving environmental, epigenetic and genetic events. Thus, the next generation of drug treatment will focus on combined therapies selective for several decisive events that characterize these disorders. Lowering the burden of protein aggregation, oxidative and nitrosative stress, mitochondrial injury, inflammatory response and heavy metal accumulation in the brain so as to re-establish neurotransmission and block excitotoxicity may prove beneficial in the treatment of several neurodegenerative diseases.

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COMPETING INTERESTS STATEMENT

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